DA04 Conditions and decisions

Week 2, Python you need | Session 4 of 24 | Python for Data Analysis

Today's question. Is this sale big enough to count as a large order?

Objective. By the end: you can write a decision that gives a different answer for True and False, and test it at the edge.

Before this session: DA03 Arithmetic, revenue and reading errors. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
comparison	A question that gives True or False, such as units >= 5.	Asking whether a pupil is tall enough to ride.
Boolean	A value that is either True or False.	A light that is on or off.
if	Run the indented lines only when the condition is True.	Enter only if you have a ticket.
else	Run these lines when the condition is False.	Otherwise, go to the queue.
indentation	Four spaces at the start of a line show which lines belong to the if.	The inside of a bracket in a sentence.
and, or	and needs both conditions True. or needs at least one.	Entry needs a ticket and an ID. A discount needs a student card or a coupon.

Block A. Compare and decide

A comparison asks a question. == asks whether two values are equal. >= means at least. The word if runs its indented lines only when the answer is True. A colon ends the if line and four spaces mark the lines that belong to it.

Predict. Predict what prints for a budget of 4999, then for 5000, then for 5001. Write your guess on paper first.

Your guess, written before you run it

```
budget = 5000
print(budget >= 5000)
if budget >= 5000:
    print("Buy")
else:
    print("Save")
```

Run it, then compare. The cell prints True and then Buy. With a budget of 4999 the comparison is False and the cell prints Save. 5000 and 5001 both print Buy, because >= includes the edge value.

Investigate.

1. Why does 5000 print Buy?
2. What is the difference between = and ==?

Modify. Set budget to 4999 and run the cell again.

Block B. Two conditions together

A large order needs 5 units or more. An expensive item costs 2000 shillings or more. and needs both to be True. or needs at least one.

Predict. Predict the answer for 6 units of an item at 1000 shillings, for both rules. Write your guess on paper first.

Your guess, written before you run it

```
units = 6
price = 1000
large = units >= 5
expensive = price >= 2000
print(large and expensive)
print(large or expensive)
if large and expensive:
    print("Large and expensive")
elif large:
    print("Large only")
else:
    print("Small")
```

Run it, then compare. The cell prints False, then True, then Large only. The order is large and not expensive, so and gives False and or gives True. elif means else if: it tests a second condition when the first is False.

Investigate.

1. Why is the answer to large and expensive False?
2. Which branch runs for 3 units at 2500 shillings?

Modify. Change the price to 2500 and run the cell again.

Make

Write a decision for a ticket that costs 5000. Print "Buy" if the budget is at least 5000 and otherwise print "Save". Test 4999, 5000 and 5001.

Save your work as DA04. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
IndentationError: expected an indented block	The line under if or else has no spaces in front of it.	Add four spaces to every line that belongs to the if.
SyntaxError: invalid syntax at if budget = 5000	A single = was used where a question needs ==.	Use == or >= to ask a question. Use = only to store a value.

Exit ticket

A comparison gives True or False, and an if runs its indented lines only when it is True.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or assessments. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been submitted until you use the route in START_HERE.pdf.